

V4.2 Toy Model 2F

Macroscopic Irreversibility and Record Stability Threshold

Rev A - Exploratory Quantum-Side Compatibility Report

Date: 2026.05.25 | Continuation after Toy Model 2E

Status: Toy Model 2F tests how decoherence becomes practically irreversible when phase information disperses into many environment records. It preserves global unitarity. It does not claim fundamental information destruction.

1. Purpose

Toy Model 2E showed that interference can be recovered from a small environment only when records are recombined and sorted correctly. Toy Model 2F asks why macroscopic objects do not show recoverable interference in practice.

The answer tested here is a scaling barrier: each extra environmental record multiplies the loss of accessible phase coherence and increases the erasure bookkeeping burden.

2. Multi-record environment model

Instead of one environment record, let the system correlate with N environment subsystems:

$$|\Psi_{SE}\rangle = \alpha |0\rangle |E_0(1)\rangle \dots |E_0(N)\rangle + \beta |1\rangle |E_1(1)\rangle \dots |E_1(N)\rangle$$

For each record subsystem k, define the overlap:

$$q_k = \langle E_1(k) | E_0(k) \rangle$$

The total environment overlap is the product:

$$\Gamma_N = \text{product over } k \text{ of } q_k$$

If all records have the same overlap q, this becomes:

$$\Gamma_N = q^N$$

3. Meaning of the formula

- q means how similar the two possible records are for one environment subsystem.
- q = 1 means the record carries no which-path information.
- q = 0 means the record perfectly distinguishes the alternatives.
- Γ_N is the remaining total overlap between the two macroscopic environment records.
- The off-diagonal interference terms of the observer-accessible state are multiplied by Γ_N .
- If $|\Gamma_N|$ becomes very small, interference is not fundamentally destroyed, but it becomes practically inaccessible.

4. Reduced density matrix

The observer-accessible reduced state becomes:

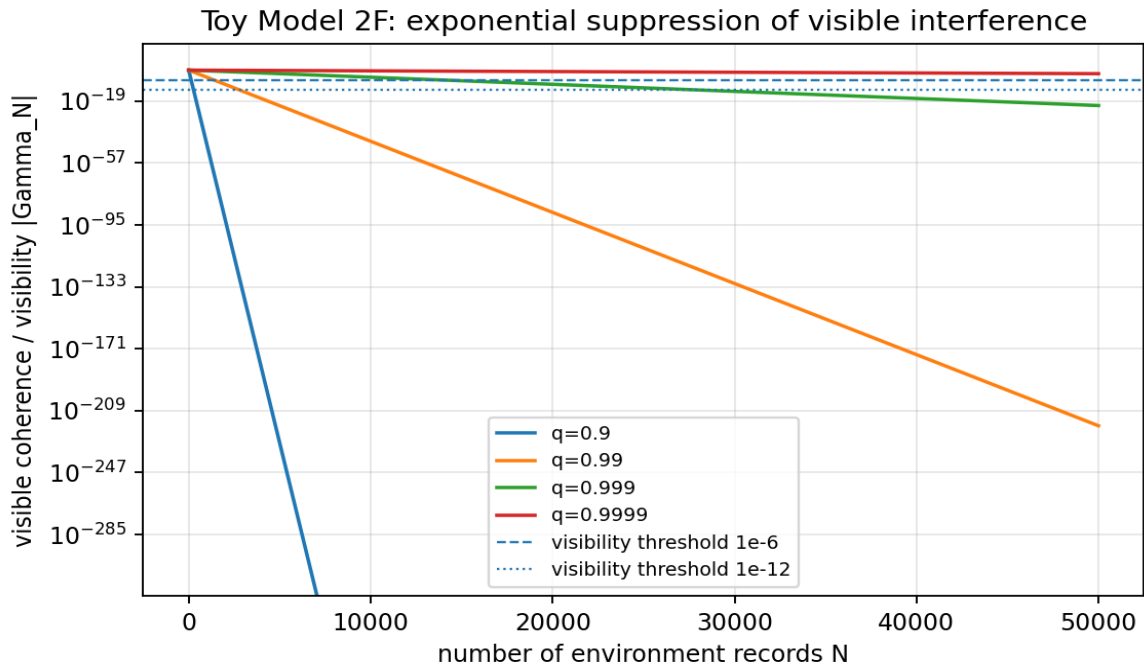
$$\rho_S = \begin{bmatrix} |\alpha|^2 & \alpha \beta^* \Gamma_N \\ \alpha^* \beta \Gamma_N^* & |\beta|^2 \end{bmatrix}$$

For a balanced state $\alpha = \beta = 1/\sqrt{2}$, the visible coherence is simply:

$$\text{visible coherence} = |\Gamma_N| = q^N$$

This is the avalanche effect: many small record correlations multiply into a very large suppression of visible interference.

5. Exponential suppression



Even when each individual record is weak, many records suppress coherence exponentially. For q slightly below 1, q^N eventually becomes tiny as N grows.

6. Record-stability thresholds

per-record q	N for $1e-3$	N for $1e-6$	N for $1e-12$	N for $1e-30$
0.9	66	132	263	656
0.99	688	1375	2750	6874
0.999	6905	13809	27618	69044
0.9999	69075	138149	276297	690741

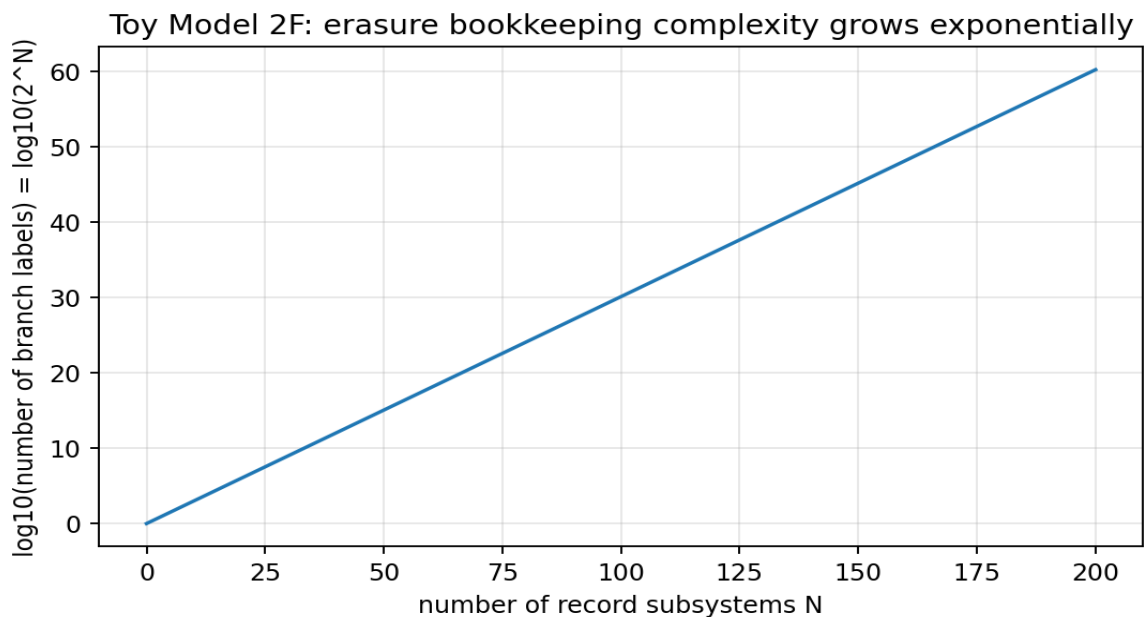
A record-stability threshold is not a new law of physics. It is a practical threshold where interference becomes too small to recover without extreme control of the environment.

7. Erasure complexity

To reverse decoherence perfectly, the erasure operation must control the relevant environment records coherently. A crude bookkeeping measure is the number of branch labels:

$$\text{number of record branches} = 2^N$$

This grows exponentially. The logarithmic plot below shows how quickly the bookkeeping burden grows.

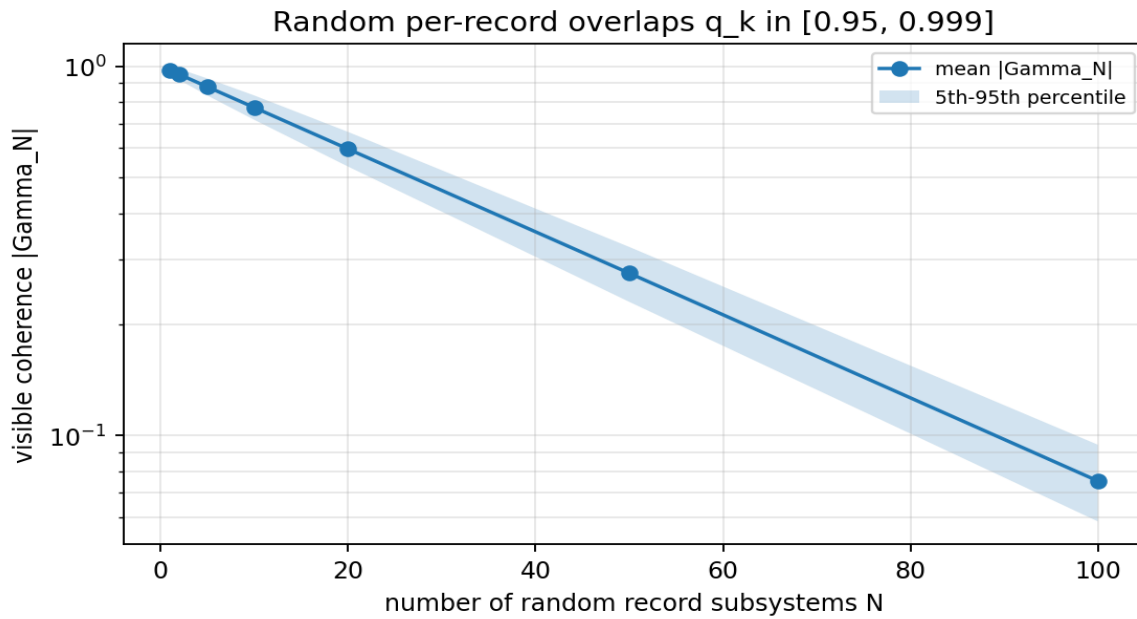


This is why microscopic erasure experiments are possible, while macroscopic erasure is not realistically available. The global state may remain unitary in principle, but the 1T observer cannot practically reverse the record avalanche.

8. Random-overlap numerical check

A Monte Carlo check used random per-record overlaps q_k in the range $[0.95, 0.999]$. The total overlap still decreases rapidly as more records accumulate.

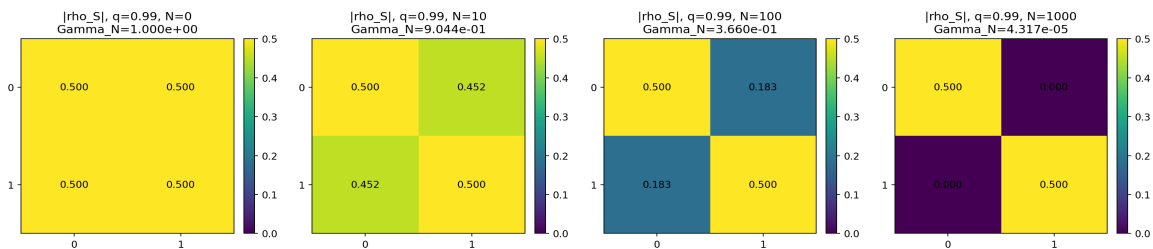
N records	mean $ \Gamma_N $	median	5th percentile	95th percentile
1	9.742138e-01	9.741339e-01	9.524746e-01	9.964241e-01
2	9.498719e-01	9.496664e-01	9.173972e-01	9.826620e-01
5	8.788240e-01	8.785963e-01	8.325169e-01	9.258997e-01
10	7.725602e-01	7.723209e-01	7.156564e-01	8.329437e-01
20	5.966666e-01	5.950485e-01	5.348763e-01	6.636812e-01
50	2.751681e-01	2.738312e-01	2.301935e-01	3.240615e-01
100	7.536883e-02	7.466490e-02	5.857233e-02	9.445141e-02



Numerical product check max error: $4.441e-16$

9. Density matrix snapshots

For $q = 0.99$, the off-diagonal terms shrink as the number of records increases. The diagonal probabilities remain stable.



10. Interpretation inside the 6D Projection Framework

Toy Model 2F supports the following limited interpretation:

- hidden relational phase information
 - > dispersed across many environment records
 - > Γ_N becomes exponentially small
 - > 1T observer sees stable classical record
 - > recovery remains possible only in principle, not practically

This matches the intended guardrail: the underlying relational state is not destroyed. Macroscopic irreversibility is a statistical and complexity barrier inside the observer-accessible projection.

11. What Toy Model 2F achieves

- It extends decoherence from one environment record to N records.
- It derives the total overlap Γ_N as a product of record overlaps.
- It shows exponential suppression of visible interference.
- It defines practical record-stability thresholds.
- It separates in-principle reversibility from practical macroscopic irreversibility.
- It preserves the guardrail that global unitarity is not destroyed.

12. What Toy Model 2F does not achieve

- It does not prove physical 3T existence.
- It does not solve the measurement problem.
- It does not derive the Born rule.
- It does not show fundamental collapse.
- It does not calculate a real laboratory decoherence rate for a specific macroscopic object.
- It does not produce a new experimental prediction yet.

13. Close-out statement for 2F

Toy Model 2F successfully formalizes macroscopic irreversibility as a record-stability threshold. When a system becomes correlated with N environment records, the accessible interference is suppressed by $\Gamma_N = \text{product}(q_k)$, or q^N for identical records. This creates exponential practical irreversibility without destroying global unitarity. In the reciprocal projection framework, stable classical records emerge because phase information is dispersed across too many environmental degrees of freedom for the 1T observer to coherently recombine. This is a compatibility result, not proof of physical 3T existence or a solution to the measurement problem.

14. Recommended next chapter

The next logical chapter is Toy Model 2G: Quantum Chapter Synthesis and Formal Guardrail Consolidation.

1. Combine 2A through 2F into one clean quantum-side close-out paper.
2. Separate what was proven, what was only made compatible, and what remains open.
3. Identify which parts are standard quantum mechanics restated in 6D language.
4. Identify the possible genuinely new structural claim.
5. Define the next hard target: deriving or explaining the origin of complex amplitudes, Hilbert space, or Born-rule weighting.

15. Rebind prompt for continuation

We are resuming after Toy Model 2F.

2F tested macroscopic irreversibility and record stability threshold.

Multi-record environment:

$$|\Psi_{SE}\rangle = \alpha |0\rangle |E_0(1)\rangle \dots |E_0(N)\rangle + \beta |1\rangle |E_1(1)\rangle \dots |E_1(N)\rangle$$

Per-record overlap:

$$q_k = \langle E_1(k) | E_0(k) \rangle$$

Total overlap:

$$\Gamma_N = \text{product}(q_k)$$

Identical records:

$$\Gamma_N = q^N$$

Reduced state:

$$\rho_S = \begin{bmatrix} |\alpha|^2 & \alpha \beta^* \Gamma_N \\ \alpha^* \beta \Gamma_N^* & |\beta|^2 \end{bmatrix}$$

Result:

Visible interference is exponentially suppressed as N grows.

Macroscopic irreversibility is practical/statistical, not fundamental destruction of global unitarity.

Guardrail:

No proof of physical 3T existence. No derivation of Born rule. No solution to measurement collapse.

Next recommended chapter:
Toy Model 2G - Quantum Chapter Synthesis and Formal Guardrail Consolidation.